STAFF REVIEW · H446 227 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. A FIFO queue.

2. Stored state belonging to that object.

3. The local variable.

4. It lets you inspect statements, parameters and field updates inside the called function.

I Do — source worked steps

1. Predict first — Step Into Add and inspect the local points and the field Points.

2. Model the reasoning — The local becomes 8 while the object field remains 3; the assignment targets the wrong variable.

3. Model the reasoning — Replace it with Points += amount; and rerun. The field should be 8.

We Do — guided CHECK

1. Model the reasoning
The method always assigns true, so the second call fails. Fix to IsOn = !IsOn; and test twice.

You Do — source mark schemes

1. [3] Step Into opens the called method for line-by-line execution [2]; Step Over executes it and stops at the next calling statement [1].

2. [5] Local variable shadows property and may be uninitialised/invalid [2]; use `this.Total += n;` [2]; verify by testing initial Total=4, Increase(3) gives 7 [1].

3. [6] Set breakpoint before call [1]; step into method [1]; inspect list count before update [1]; step through every append [1]; find duplicate append or duplicate caller invocation [1]; fix and retest one append [1].

Exit Ticket — source answer

1. False. The call executes; the debugger does not stop inside it.